

Andreas Tzitzikas

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Education

University of Maryland , College Park, MD	Expected May 2027
<i>Master of Science in Computer Engineering</i>	GPA: 3.9
<i>Bachelor of Science in Computer Engineering</i> (GPA: 3.89)	Expected May 2026
Relevant Coursework: Parallel Computing (CMSC416), Algorithms (CMSC351), Organization of Programming Languages (CMSC330), Computer Systems (CMSC216), Operating Systems (ENEE447), Applied Stat (STAT400)	

Experience

Embedded Systems Intern , Alchemity – College Park, MD	Jun 2025 – Aug 2025
- Developed a multi-threaded test automation framework in Rust, reducing validation time by 70% through parallelized data acquisition and error logging. - Built end-to-end control interfaces including desktop GUIs, CLI tools, and embedded communication protocols to streamline workflows between hardware and software teams. - Implemented real-time data processing and visualization systems with performance monitoring, error handling, and user-friendly interfaces for experimental control.	

Projects

CPU Performance Analysis Framework Python, Bash, Automation	Aug 2025 - Present
- Developed comprehensive benchmarking suite in Python for automated performance analysis, implementing statistical analysis and workload characterization across different CPU architectures. - Architected modular analysis framework with custom scripts for IPC measurement, performance profiling, and automated report generation with detailed trade-off analysis. - Built scalable testing infrastructure supporting parallel execution and data visualization for comparative architecture studies and optimization research.	
Parallel Game of Life Implementation MPI, C/C++, Performance Optimization	Jan 2025 – May 2025
- Implemented high-performance parallel version of Conway's Game of Life using MPI with 1D row decomposition, achieving scalable performance across 128 processes with 2x reference throughput. - Architected non-blocking Isend/Irecv communication patterns with ghost cell synchronization and dynamic load balancing for optimal parallel efficiency. - Developed comprehensive timing analysis framework using MPI_Wtime for performance characterization and bottleneck identification in distributed computing environments.	
SmallC Compiler Development OCaml, Compiler Theory	Jan 2025 – May 2025
- Built complete SmallC compiler frontend with lexer, parser, and AST evaluator, implementing recursive descent parsing and tree-walking interpretation for C subset. - Developed AST optimization engine performing constant folding, constant propagation, branch folding, and loop optimization to minimize runtime computation. - Implemented type checker that verifies expressions are well-typed before evaluation, preventing runtime type errors through comprehensive static analysis. - Created type inferencer that determines variable types from Unknown_Type declarations, enabling flexible type handling while maintaining type safety guarantees.	
Space Invaders Video Game Java, Object-Oriented Design, AWT Graphics	Jan 2025 – May 2025
- Designed complete Space Invaders game using advanced OOP principles including inheritance, polymorphism, inner classes, and interfaces with lambda expressions for bonus systems. - Implemented real-time graphics engine with collision detection algorithms, coordinated enemy movement patterns, and progressive difficulty scaling using prime number-based firing rates. - Architected modular game architecture with 15 enemies, boss battles, scoring mechanics, and complete win/lose conditions demonstrating comprehensive software design patterns.	

Skills

Programming Languages: Rust, Python, C/C++, Java, OCaml, Bash, MPI, OpenMP, CUDA
Software Development: Object-Oriented Design, Algorithms, Data Structures, Parallel Computing, Git, Linux
Frameworks & Tools: MPI/OpenMP/CUDA, JUnit Testing, Build Automation, Version Control, Debugging, Profiling
Methodologies: Agile Development, Test-Driven Development, Code Review, Documentation, Algorithm Analysis